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# Breaking the Regional Barrier: Inductive Semantic Topology Learning for Worldwide Air Quality



*NeurIPS 2026 Conference Submission1773 Authors*

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👁 Conference, Senior Area Chairs, Area Chairs, Reviewers, Authors

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**TL;DR:** The first global-scale station-level air quality forecasting framework, OmniAir, achieving 10x speedup.

## Abstract:

Global air quality forecasting grapples with extreme spatial heterogeneity and the poor generalization of existing transductive models to unseen regions. To tackle this, we propose **OmniAir**, a semantic topology learning framework tailored for global station-level prediction. By encoding invariant physical environmental attributes into generalizable station identities and dynamically constructing adaptive sparse topologies, our approach effectively captures long-range non-Euclidean correlations and physical diffusion patterns across unevenly distributed global networks. We further curate **WorldAir**, a massive dataset covering over 7,800 stations worldwide. Extensive experiments show that OmniAir achieves state-of-the-art performance against 18 baselines, maintaining high efficiency and scalability with speeds nearly 10 times faster than existing models, while effectively bridging the monitoring gap in data-sparse regions. Our code is available at <https://anonymous.4open.science/r/OmniAir-9B40/> (<https://anonymous.4open.science/r/OmniAir-9B40/>).

**Primary Area:** AI/ML for physical sciences (e.g., physics, chemistry, Earth sciences, astronomy)

**Secondary Area:** AI and network science (e.g., graph neural networks, communication)

**Contribution Type:** General: Most submissions will fall into this type.

**Submission Number:** 1773

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📅 09 Jun 2026, 09:26 👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewer yFPm

**Summary:** The paper "Breaking the Regional Barrier: Inductive Semantic Topology Learning for Worldwide Air Quality Forecasting" introduces OmniAir, a semantic topology learning framework designed for global station-level air

quality prediction. To overcome the poor generalization of existing transductive models to unseen regions, OmniAir encodes invariant environmental attributes into generalizable station identities and dynamically constructs adaptive sparse topologies. This allows the model to capture non-Euclidean correlations and physical diffusion patterns across globally distributed networks. The authors also curate and release WorldAir, a massive dataset covering over 7,800 stations. Experiments demonstrate that OmniAir achieves state-of-the-art results against 18 baselines while remaining highly scalable and capable of inductive forecasting in data-sparse regions.

**Contribution Type:** general

**Strengths And Weaknesses:** Strengths:

Inductive Generalization: Shifting from transductive learning to invariant physical attributes effectively addresses the generalization gap for unseen monitoring stations, a critical bottleneck in global spatio-temporal forecasting.

Scalability and Efficiency: The dynamic sparse topology approach scales well, reportedly running significantly faster than dense graph-based counterparts without sacrificing predictive power.

Significant Resource Contribution: The introduction of the WorldAir dataset is a substantial asset for the community, providing a much-needed benchmark for global-scale air quality models.

Weaknesses:

Topological Stability: Dynamic sparse topologies can sometimes lead to training instability or structural jitter in continuous time series. It is unclear how the model enforces temporal consistency in the graph structure.

Baseline Context: While outperforming 18 baselines is impressive, ensuring these baselines include the most recent spatio-temporal foundation models is necessary for a definitive state-of-the-art claim.

**Quality:** 4

**Clarity:** 3

**Significance:** 4

**Originality:** 4

**Questions:** How does the model balance the weighting between the invariant physical environmental attributes and dynamic temporal features during the topology construction?

Are there specific regularization techniques applied to the dynamic sparse topologies to prevent structural collapse or excessive jitter across consecutive time steps?

How does OmniAir handle extreme, out-of-distribution anomalous events (e.g., severe wildfires or industrial accidents) compared to standard seasonal fluctuations?

**Limitations:** Yes. The authors successfully address the challenges of transductive models, but further discussion on the computational overhead of dynamic topology construction during inference, or potential failure modes in regions with highly unique microclimates, would strengthen the work.

**Rating:** 5

**Confidence:** 2

**Ethical Concerns:** NO or VERY MINOR ethics concerns only

**Paper Formatting Concerns:** None Observed

**Code Of Conduct Acknowledgement:** Yes

**Responsible Reviewing Acknowledgement:** Yes

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